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Electrospun PTFE coated stent and method of use

Abstract

A stent or other prosthesis may be formed by coating a single continuous wire scaffold with a polymer coating. The polymer coating may consist of layers of electrospun polytetrafluoroethylene (PTFE). Electrospun PTFE of certain porosities may permit endothelial cell growth within the prosthesis.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 13/360,444, filed on Jan. 27, 2012 and titled, “Electrospun PTFE Coated Stent and Methods of Use,” which claims priority to U.S. Provisional Application No. 61/437,404, filed on Jan. 28, 2011, and titled “Electrospun PTFE Coated Stent and Method of Use,” both of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure relates generally to medical devices. More specifically, the present disclosure relates to stents or other prostheses, particularly prosthesis coated by electrospun polytetrafluoroethylene (PTFE).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The embodiments disclosed herein will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. These drawings depict only typical embodiments, which will be described with additional specificity and detail through use of the accompanying drawings in which:
- (2) FIG. 1 is a front elevation view of one embodiment of a stent.
 - (3) FIG. 2A is a cutaway view of the front portions of two adjacent coils of one embodiment of a stent.
 - (4) FIG. 2B is a detail view of one end of the stent of FIG. 1.
 - (5) FIG. 2C is a detail view of an alternative design of an end portion of a stent.
 - (6) FIG. 3 is a top view of another embodiment of a stent with flared ends.
 - (7) FIG. 4 is a front elevation view of the stent of FIG. 3.
 - (8) FIG. 5 is a perspective view of one embodiment of a stent, illustrating how a wire may be shaped to form the structure of the stent.

- (9) FIG. 6A is a perspective view of a covered stent.
- (10) FIG. 6B is a cross sectional view of the stent of FIG. 6A along the plane 6B-6B.
- (11) FIG. 7 illustrates one embodiment of a stent deployed in a body lumen.
- (12) FIGS. 8A-8D are scanning electron micrograph (“SEM”) images of one embodiment of an electrospun PTFE outer covering for a stent.
- (13) FIGS. 9A-9D are SEM images of an electrospun PTFE inner layer of the covering of the stent of FIGS. 8A-8D.
- (14) FIGS. 10A-10D are SEM images of an electrospun PTFE outer covering of another embodiment of a stent.
- (15) FIGS. 11A-11D are SEM images of an electrospun PTFE inner layer of the covering of the stent of FIGS. 10A-10D.

DETAILED DESCRIPTION

- (16) Stents may be deployed in various body lumens for a variety of purposes. Stents may be deployed, for example, in the central venous system for a variety of therapeutic purposes including the treatment of occlusions within the lumens of that system. It will be appreciated that the current disclosure may be applicable to stents designed for the central venous (“CV”) system, peripheral vascular (“PV”) stents, abdominal aortic aneurism (“AAA”) stents, bronchial stents, esophageal stents, biliary stents, or any other stent. Further, the present disclosure may equally be applicable to other prosthesis such as grafts. Thus, the disclosure provided below in connection with specific examples of stents may apply analogously to other prostheses.
- (17) It will be readily understood that the components of the embodiments as generally described and illustrated in the Figures herein could be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of various embodiments, as represented in the Figures, is not intended to limit the scope of the disclosure, but is merely representative of various embodiments. While the various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.
- (18) The phrases “connected to,” “coupled to,” and “in communication with” refer to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be coupled to each other even though they are not in direct contact with each other. For example, two components may be coupled to each other through an intermediate component.
- (19) The directional terms “proximal” and “distal” are used herein to refer to opposite locations on a stent. The proximal end of a stent is defined as the end of the stent closest to the practitioner when the stent is disposed within a deployment device which is being used by the practitioner. The distal end is the end opposite the proximal end, along the longitudinal direction of the stent, or the end furthest from the practitioner. It is understood that, as used in the art, these terms may have different meanings once the stent is deployed (i.e. the “proximal” end may refer to the end closest to the head or heart of the patient depending on application). For consistency, as used herein, the ends of the stent labeled “proximal” and “distal” prior to deployment remain the same regardless of whether the stent is deployed. The longitudinal direction of the stent is the direction along the axis of a generally tubular stent. In embodiments where a stent is composed of a metal wire structure coupled to a polymer layer, the metal structure is referred to as the “scaffolding” and the polymer layer as the “coating.” The term “coating” may refer to a single layer of polymer, multiple layers of the same polymer, or layers comprising distinct polymers used in combination.
- (20) Lumens within the central venous system are generally lined with endothelial cells. This lining of endothelial cells throughout the central venous system makes up the endothelium. The endothelium acts as an interface between blood flowing through the lumens of the central venous system and the inner walls of the lumens. The endothelium, among other functions, reduces or prevents turbulent blood flow within the lumen.
- (21) A therapeutic stent which includes a coating of porous or semi-porous material may permit the

formation of an endothelial layer on the inside surface of the stent. A stent which permits the formation of the endothelium within the stent may further promote healing at the therapeutic region. For example, a stent coated with endothelial cells may be more consistent with the surrounding body lumens, thereby resulting in less turbulent blood flow or a decreased risk of thrombosis, or the formation of blood clots. A stent which permits the formation of an endothelial layer on the inside surface of the stent may therefore be particularly biocompatible, resulting in less trauma at the point of application and fewer side effects.

(22) Electrospun polytetrafluoroethylene (PTFE) may be used as a stent coating where endothelial cell growth is desirable. “Electrospinning” refers to a process for forming mats, tubes, or other shapes by depositing small strings of PTFE on charged surfaces. The electrospinning process controls the thickness, density, porosity, and other characteristics of the PTFE so formed.

Electrospinning of PTFE is described in United States Patent Application, Publication No. US 2010/0193999, which is incorporated herein by reference.

(23) The present disclosure relates to a stent which has, in certain embodiments, metal scaffolding coated with at least one layer of electrospun PTFE. It will be appreciated that, though particular structures and coatings are described below, any feature of the scaffolding or coating described below may be combined with any other disclosed feature without departing from the scope of the current disclosure. For example, certain figures show metal scaffolding without any coating; the features described and illustrated in those figures may be combined with any combination of coatings disclosed herein.

(24) FIGS. 1, 2A, and 2B show views of a possible embodiment of a stent. FIGS. 3 and 4 are views of one embodiment of a stent which includes flared ends. FIG. 5 illustrates one embodiment of how a wire may be shaped to form a scaffold for a stent. FIGS. 6A and 6B illustrate an embodiment of a covered stent. FIG. 7 illustrates a stent deployed within a body lumen. Finally, FIGS. 8A-11D are scanning electron micrographs (SEMs) of possible coatings for stents. As indicated above, it will be understood that, regardless of whether the stent illustrated in any particular figure is illustrated with a particular coating, or any coating at all, any embodiment may be configured with any of the combinations of coatings shown or described herein.

(25) FIG. 1 illustrates a front elevation view of an embodiment of a stent **100**. The illustrated embodiment depicts one embodiment of a configuration for a metal wire **110** forming a scaffolding structure. As depicted in FIG. 1, the scaffolding may consist of a single continuous wire.

(26) Referring generally to FIGS. 1, 2A, and 2B, particular features of the illustrated stent are indicated. It will be appreciated that the numerals and designations used in any figure apply to analogous features in other illustrated embodiments, whether or not the feature is so identified in each figure. As generally shown in these Figures, the stent **100** may consist of a wire **110** shaped to form scaffolding. The wire **110** may be shaped in a wave-type configuration, the waves defining apexes **102** and arms **104** of the stent. The scaffolding may further be coupled to a covering layer (not pictured). Additionally, in some embodiments, any covering as disclosed herein may be applied to any type of scaffolding or stent frame, for example, laser cut stent frames, polymeric stent frames, wire scaffolding, and so forth.

(27) The stent **100** may be designed such that the midsection of the stent is “harder” than the ends. The “hardness” of the stent refers to the relative strength of the stent (e.g., its compressibility). A harder portion of the stent will have greater strength (i.e. exert a greater radial outward force) than a softer portion. In one embodiment, the midsection of the stent is harder than the proximal and distal end sections which are relatively softer.

(28) Four basic design parameters may be manipulated to influence the properties (hardness, strength, crush force, hoop force, flexibility, etc.) of the illustrated stent. These properties are: (1) apex to apex distance, designated as H.sub.x in FIGS. 1 and 2A; (2) arm length, designated as A.sub.x in FIGS. 1 and 2A; (3) apex radius, designated as R.sub.x in FIG. 2A; and (4) the diameter of the scaffolding wire **110**. These values may or may not be constant at different points on a stent.

Thus, the subscript “x” is used generically; that is, each distance identified as “H” refers to an apex to apex distance with subscripts 1, 2, 3, etc., signifying the apex to apex distance at a particular point. It will be appreciated that these subscript designations do not necessarily refer to a specific distance, but may be used relatively (i.e., H.sub.1 may be designated as smaller than H.sub.2 without assigning any precise value to either measurement). Further, as will be apparent to one skilled in the art having the benefit of this disclosure, an analogous pattern of measurements and subscripts is employed for other parameters described herein, for example A.sub.x and R.sub.x.

(29) The overall stent design may be configured to optimize desired radial force, crush profile, and strain profile. The stent design parameters may each be configured and tuned to create desired stent characteristics. For example, the strain profile may be configured to be less than the failure point for the material being used.

(30) A first parameter, the apex to apex distance, is designated as H.sub.x. This measurement signifies the distance between a first apex and a second apex where both apices substantially lie along a line on the outside diameter of the stent which is co-planar with, and parallel to, the longitudinal axis of the stent. In some embodiments, H.sub.x may be constant along the entire length of the stent. In other embodiments the length of the stent may be divided into one or more “zones” where H.sub.x is constant within a zone, but each zone may have a different H.sub.x. In still other embodiments H.sub.x may vary along the entire length of the stent. H.sub.x may be configured, in connection with the other design parameters, to determine the properties of the stent. Generally, regions of the stent with a smaller H.sub.x value will be harder than regions with a larger H.sub.x value.

(31) In the embodiment illustrated in FIG. 1, there are two “flare zones” at either end of the stent and a midbody zone along the remaining length of the stent. In the illustrated embodiment, H.sub.1 designates the apex to apex distance in the midbody zone of the stent and H.sub.2 designates the apex to apex distance in the flare zones of the stent. In the illustrated embodiment, the apex to apex distance, H.sub.2, is the same in both the flare zone near the distal end of the stent and the flare zone near the proximal end of the stent. In some embodiments H.sub.1 may be smaller than H.sub.2, resulting in a stent that is relatively harder in the midbody and relatively softer on the ends. A stent with such properties may be utilized in applications where strength is necessary along the midbody, for example to treat a tumor or other occlusion, but the ends are configured to rest on healthy tissue where softer ends will minimize trauma to the healthy tissue.

(32) In embodiments where soft ends and a hard midbody are desirable, H.sub.1 may be between about 2 mm and 30 mm and H.sub.2 between about 2.1 mm and 30.1 mm. For example, in stents for CV or PV application, H.sub.1 may be between about 3 mm and 10 mm and H.sub.2 between about 3.1 mm and 10.1 mm, such as: $3\text{ mm} < \text{H.sub.1} < 8\text{ mm}$ and $3.5\text{ mm} < \text{H.sub.2} < 9\text{ mm}$; $3\text{ mm} < \text{H.sub.1} < 6.5\text{ mm}$ and $4\text{ mm} < \text{H.sub.2} < 8\text{ mm}$; or $3\text{ mm} < \text{H.sub.1} < 5\text{ mm}$ and $5.5\text{ mm} < \text{H.sub.2} < 6.5\text{ mm}$.

(33) In other embodiments where two or more apex to apex lengths are present in one stent, the change in apex to apex length may be correlated to the displacement of the apices from the midpoint of the stent. In other words, the apex to apex length may increase incrementally as one moves away from the midpoint of the stent toward the ends in a manner that gives the stent the same geometry, and therefore the same properties, on either side of the midpoint of the length of the stent. In other embodiments, different geometries may be utilized at any point along the length of the stent. It will be appreciated that the ranges of values for H.sub.x disclosed above apply analogously to embodiments where the stent has multiple apex to apex lengths. For example, in one embodiment a stent may have an apex to apex length at midbody within one of the ranges disclosed above for H.sub.1, and the value of H.sub.x may vary incrementally, in steps, or some other pattern, along the length of the stent reaching an apex to apex length at the ends within the complimentary range for H.sub.2.

(34) Moreover, in some embodiments, the value of H.sub.x may be small enough that adjacent coils

are “nested” within each other. In other words, the apexes of a first helical coil may extend up into the spaces just below the apexes of the next adjacent coil. In other words, apexes of lower coils may extend a sufficient amount so as to be disposed between the arms of higher coils. In other embodiments the value of H.sub.x may be large enough that adjacent coils are completely separated. In embodiments wherein adjacent coils are “nested,” the number of wires at any particular cross section of the stent may be higher than a non-nested stent. In other words, cutting the stent along an imaginary plane disposed orthogonal to the longitudinal axis of the stent will intersect more wires if the stent is nested as compared to non-nested stents. The smaller the value of H.sub.x, the more rows may be intersected by such a plane (that is, more than just the next adjacent row may extend into the spaces below the apexes of a particular row). Nested stents may create relatively higher strains in the scaffolding structure when the stent is loaded into a delivery catheter. In some instances the delivery catheter for a nested stent may therefore be relatively larger than a delivery catheter configured for a non-nested stent. Further, nested stents may be relatively stiff as compared to non-nested stents with similar parameters.

(35) As will be apparent to those skilled in the art having the benefit of this disclosure, stents with a hard mid body and soft ends may be desirable for a variety of applications. Further, in some instances a basically “symmetric” stent may be desirable; in other words, a stent with certain properties at the midbody section and other properties at the ends, where the properties at both ends are substantially identical. Of course, other embodiments may have varied properties along the entire length of the stent. It will be appreciated that while the effect of changing variables, for instance the difference between H.sub.1 and H.sub.2, may be described in connection with a substantially symmetric stent (as in FIG. 1) the same principles may be utilized to control the properties of a stent where the geometry varies along the entire length of the stent. As will be appreciated by those skilled in the art having the benefit of this disclosure, this applies to each of the variable parameters described herein, for example H.sub.x, A.sub.x, and R.sub.x.

(36) A second parameter, arm length, is designated as A.sub.x in FIGS. 1 and 2A. As with H.sub.x, A.sub.x may be constant along the length of the stent, be constant within zones, or vary along the length of the stent. Variations in the length of A.sub.x may be configured in conjunction with variations in the other parameters to create a stent with a particular set of properties. Generally, regions of the stent where A.sub.x is relatively shorter will be harder than regions where A.sub.x is longer.

(37) In some embodiments, the arm length A.sub.1 near the midsection of the stent **100** will be shorter than the arm length A.sub.2 near the ends. This configuration may result in the stent being relatively harder in the midsection. In embodiments where soft ends and a hard midbody are desirable, A.sub.1 may be between about 2 mm and 30 mm and A.sub.2 between about 2.1 mm and 30.1 mm. For example, in stents for CV or PV application, A.sub.1 may be between about 2 mm and 10 mm and A.sub.2 between about 2.1 mm and 10.1 mm, such as: $2.5\text{ mm} < A_{\text{sub.1}} < 8\text{ mm}$ and $3\text{ mm} < A_{\text{sub.2}} < 9\text{ mm}$; $3\text{ mm} < A_{\text{sub.1}} < 6\text{ mm}$ and $4\text{ mm} < A_{\text{sub.2}} < 7.5\text{ mm}$; or $4\text{ mm} < A_{\text{sub.1}} < 5\text{ mm}$ and $5\text{ mm} < A_{\text{sub.2}} < 6\text{ mm}$.

(38) In other embodiments where two or more arm lengths are present in one stent, the change in arm length may be correlated to the displacement of the arm from the midpoint along of the stent. In other words, the arm length may increase incrementally as one moves away from the midpoint of the stent toward the ends in a manner that gives the stent the same geometry, and therefore the same properties, on either side of the midpoint of the length of the stent. In other embodiments, different geometries may be utilized at any point along the length of the stent. It will be appreciated that the ranges of values for A.sub.x disclosed above apply analogously to embodiments where the stent has multiple arm lengths. For example, in one embodiment a stent may have an arm length at midbody within one of the ranges disclosed above for A.sub.1, and the value of A.sub.x may vary incrementally, in steps, or some other pattern, along the length of the stent reaching an arm length at the ends within the complimentary range for A.sub.2.

(39) A third parameter, the apex radius, is designated as R.sub.1 in FIG. 2A. As with H.sub.x, and A.sub.x, R.sub.x may be configured in order to create desired properties in a stent. In some embodiments, the inside radius of each apex may form an arc with has a substantially constant radius. As shown by a dashed line in FIG. 2A, this arc can be extended to form a circle within the apex. The measurement R.sub.x refers to the radius of the arc and circle so described. Further, in some embodiments the arms and apexes of the stent scaffolding are formed by molding a wire around pins protruding from a mandrel. The radius of the pin used gives the apex its shape and therefore has substantially the same radius as the apex. In some embodiments R.sub.x will be constant along the entire length of the stent, be constant within zones along the length of the stent, or vary along the entire length of the stent. Variations in the magnitude of R.sub.x may be configured in conjunction with variations in the other parameters to create a stent with a particular set of properties. Generally, regions of the stent where R.sub.x is relatively smaller will be harder than regions where R.sub.x is larger.

(40) Furthermore, in some instances, smaller values of R.sub.x may result in relatively lower strain in the wire scaffolding when the scaffolding is compressed, for example when the stent is disposed within a delivery catheter. Moreover, wires of relatively larger diameters may result in relatively lower strain at or adjacent the radius measured by R.sub.x when compressed, as compared to wires of smaller diameters. Thus, in some instances, the strain may be optimized for a particular design by varying the value of R.sub.x and the diameter of the wire forming the scaffolding.

(41) Like the other variables, R.sub.x may take on a range of values depending on the application and the desired properties of the stent. In some embodiments R.sub.x may be between about 0.25 mm and 1.5 mm. For example, in stents for CV or PV application, R.sub.x may be between about 0.35 mm and 0.70 mm, such as: $0.35\text{ mm} < R_{\text{sub}.x} < 0.65\text{ mm}$; $0.35\text{ mm} < R_{\text{sub}.x} < 0.6\text{ mm}$; or $0.4\text{ mm} < R_{\text{sub}.x} < 0.5\text{ mm}$.

(42) It will be appreciated that the disclosed ranges for R.sub.x apply whether the value of R.sub.x is constant along the length of the stent, whether the stent is divided into zones with different R.sub.x values, or whether R.sub.x varies along the entire length of the stent.

(43) The fourth parameter, wire diameter, is discussed in detail in connection with FIG. 5 below.

(44) FIG. 2A illustrates a cutaway view of the front portions of two adjacent coils of a stent. The portions of the coils depicted are meant to be illustrative, providing a clear view of the three parameters H.sub.x, A.sub.x, and R.sub.x. It will be appreciated that all three of these parameters may be configured in order to create a stent with particular properties. Any combination of the values, ranges, or relative magnitudes of these parameters disclosed herein may be used within the scope of this disclosure. As an example of these values taken together, in one embodiment of a CV or PV stent with a relatively hard midbody and softer ends, H.sub.1 may be about 4 mm and H.sub.2 about 5.9 mm; A.sub.1 may be about 4.5 mm and A.sub.2 about 5.6 mm; and R.sub.1 about 0.5 mm.

(45) FIG. 2B is a close up view of one end of a stent. In embodiments where the scaffolding is formed by a single continuous wire, FIG. 2B illustrates one way in which the end of the wire **106** may be coupled to the scaffolding. As illustrated, the wire may be disposed such that the final coil approaches and runs substantially parallel to the previous coil. This configuration results in the apex to apex distance between the two coils decreasing near the end **106** of the wire. In some embodiments this transition will occur along the distance of about 4-8 apexes along the length of the wire. For example, if a stent is configured with a apex to apex spacing of H.sub.2' along the region of the stent nearest the ends, the apex to apex distance will decrease from H.sub.2' to a smaller distance which allows the end of the wire **106** to meet the prior coil (as illustrated in FIG. 2B) over the course of about 4-8 apexes.

(46) FIG. 2C illustrates an alternative configuration of a wire scaffolding. In the embodiment of FIG. 2C, apexes **102'** alternate in relative height along the length of the wire. In particular, in the embodiment shown, the apexes form a pattern comprising a higher apex, a shorter apex, a higher

apex, a shorter apex, and so on, around the helical coil. In some instances, a stent may be configured with alternating apices at one or both ends of the stent. For example, a stent as shown in FIG. 1 may be configured with the pattern of apices **102'** and arms **104'** shown in FIG. 2C at one or both ends of the stent. Such an alternating pattern of apices may distribute the force along the vessel wall at the ends of the stent, thus creating relatively a-traumatic ends.

(47) The end **106** may be attached to the scaffolding in a variety of ways known in the art. The end **106** may be laser welded to the scaffolding or mechanically crimped to the scaffolding. In embodiments where the stent includes a polymer cover, the end **106** may be secured by simply being bound to the cover. In other instances, a string may be used to bind or tie the end **106** to adjacent portions of the scaffolding. Similarly, in some instances, a radiopaque marker may be crimped around the end **106** in such a manner as to couple the end **106** to the scaffolding. Additionally other methods known in the art may be utilized.

(48) Furthermore, in some embodiments the stent **100** may be configured with radiopaque markers at one or more points along the stent **100**. Such markers may be crimped to the scaffolding structure. In other embodiments a radiopaque ribbon, for example a gold ribbon, may be threaded or applied to the stent **100**. In some embodiments these markers may be located at or adjacent one or both ends of the stent **100**. Any radiopaque material may be used, for example gold or tantalum.

(49) Referring again to FIG. 1 as well as to FIGS. 3 and 4, the stent **100** may be configured with flared ends. It will be appreciated that in certain embodiments a stent may have a flare at both the proximal and distal ends, only at the proximal end or only at the distal end, or at neither end. In certain of these embodiments the stent **100** may have a substantially constant diameter in the midbody zone of the stent, with the ends flaring outward to a larger diameter at the ends. It will be appreciated that the geometry of the flares at the proximal and distal ends may or may not be the same.

(50) In the embodiment illustrated in FIG. 1, the stent **100** has a diameter, $D_{sub.1}$, at the midbody of the stent. This diameter may be constant along the entire midbody of the stent. The illustrated embodiment has a second diameter, $D_{sub.2}$, at the ends. This change in diameter creates a “flare zone” at the end of the stent, or an area in which the diameter is increasing and the stent therefore may be described as including a “flared” portion. In some embodiments the flare zone will be from about 1 mm to 60 mm in length. For example in certain stents designed for CV or PV application the flare zone may be from about 3 mm to about 25 mm in length, such as: from about 4 mm to 15 mm, or from about 5 mm to about 10 mm in length.

(51) FIGS. 3 and 4 also illustrate how a stent may be flared at the ends. Diameters $D_{sub.1}'$ and $D_{sub.1}''$ refer to midbody diameters, analogous to $D_{sub.1}$, while $D_{sub.2}'$ and $D_{sub.2}''$ refer to end diameters analogous to $D_{sub.2}$. Further, as illustrated in FIG. 4, the flared end may create an angle, α , between the surface of the stent at the midbody and the surface of the flare. In some instances the flare section will uniformly flare out at a constant angle, as illustrated in FIG. 4. In some embodiments angle α will be from about 1 degree to about 30 degrees. For example, in some stents designed for CV or PV application, α will be from about 2 degrees to 8 degrees, such as: from about 2.5 degrees to about 7 degrees or from about 3 degrees to about 5 degrees. In one exemplary embodiment α may be about 3.6 degrees.

(52) The stent **100** of FIG. 1 also has a length L . It will be appreciated that this length can vary depending on the desired application of the stent. In embodiments where the stent has flare zones at the ends, longer stents may or may not have proportionally longer flare zones. In some embodiments, this flare zone may be any length described above, regardless of the overall length of the stent.

(53) It will be appreciated that the disclosed stent may be formed in a variety of sizes. In some embodiments, L may be from about 20 mm to about 200 mm. For example, in CV applications the stent may have a length, L , of from about 40 mm to 100 mm or any value between, for example, at least about 50 mm, 60 mm, 70 mm, 80 mm, or 90 mm. In PV applications the stent may have a

length, L, of from about 25 mm to 150 mm or any value between, for example at least about 50 mm, 75 mm, 100 mm or 125 mm. The stent may also be longer or shorter than these exemplary values in other stent applications.

(54) Likewise the stent may be formed with a variety of diameters. In some embodiments the midbody diameter of the stent may be from about 4 mm to about 40 mm. For example, in CV or PV applications the stent may have a midbody inside diameter of about 3 mm to 16 mm or any distance within this range such as between about 5 mm to 14 mm or between about 7 mm to about 10 mm.

(55) The stent may or may not be configured with flared ends regardless of the midbody diameter employed. In some central venous embodiments the maximum diameter at the flared end will be between about 0.5 mm to about 2.5 mm greater than the midbody diameter. For example, the maximum diameter at the flared end may be between about 1 mm to about 2 mm, or alternatively between about 1.25 mm and about 1.5 mm, such as about 1.25 mm or about 1.5 mm greater than the midbody diameter.

(56) Referring now to FIG. 5, the scaffolding of the stent may be formed from a single continuous wire. In some embodiments the wire may be comprised of Nitinol (ASTM F2063), or other suitable materials. In some embodiments the wire will have a diameter between about 0.005 inches and about 0.020 inches. For example, in some stents designed for CV or PV application, the wire diameter may be from about 0.008 inches to about 0.012 inches in diameter including certain embodiments where the wire is from about 0.009 inches to about 0.011 inches in diameter or embodiments where the wire is about 0.010 inches in diameter. Furthermore stents configured for the thoracic aorta may be formed of wires up to 0.020 inches in diameter, including wires up to about 0.015 inches or 0.010 inches in diameter.

(57) FIG. 5 illustrates how, in some embodiments, the wire **110** may be wound in a helical pattern creating coils that incline along the length of the stent. The waves of the wire which form the arms and apexes may be centered around this helix, represented by the dashed line **120**.

(58) Referring now to FIGS. 6A and 6B, in some embodiments the stent **100** may be comprised of a wire **110** which forms the scaffolding and a cover **200** coupled to the scaffolding. In some embodiments this cover may be comprised of a single layer, while in other embodiments it may be comprised of 2, 3, or more layers of material. One or more layers may be comprised of a polymer.

(59) The illustrated embodiment has two cover layers, an outer layer **210** and an inner layer **220**. Portions of the scaffolding may protrude through one or both layers at certain points, or, the scaffolding may be completely enclosed on the outside diameter by the outer layer **210** and on the inside diameter by the inner layer **220**.

(60) In some embodiments the outer layer **210**, the inner layer **220**, or both may be comprised of electrospun PTFE. Electrospun PTFE consists of tubes, mats, or other shapes of PTFE formed from randomly deposited strings of PTFE. As previously indicated, electrospinning of PTFE is described in United States Patent Application, Publication No. US 2010/0193999. As described in the reference, electrospinning may comprise depositing a polymer on a collection surface, in the presence of an electrostatic field. In some instances the polymer may be electrostatically charged and may be discharged through one or more orifices.

(61) Further information, which is unique to this disclosure, relative to electrospinning PTFE or other polymer is included below. The properties of electrospun PTFE, including density and porosity, may be controlled or influenced during the creation of the electrospun PTFE, through controlling the electrospinning process.

(62) In some embodiments, a PTFE dispersion may be discharged through an orifice to electrospin the PTFE. Furthermore, in some exemplary embodiments polyethylene oxide (PEO) may be added to the PTFE dispersion prior to electrospinning the material. The PEO may be added as a fiberizing agent, to aid in the formation of PTFE fibers within the dispersion or during the process of electrospinning the material. In some instances the PEO may more readily dissolve in the PTFE

dispersion if the PEO is first mixed with water. In some examples this increased solubility may reduce the time needed to dissolve PEO in a PTFE dispersion from as long as multiple days to as little as 30 minutes. After the material is electrospun onto a collector, the material may then be sintered as further described below. In some instances the sintering process will tend to set or harden the structure of the PTFE. Furthermore, sintering may also eliminate the water and PEO, resulting in a mat of substantially pure PTFE.

(63) In one exemplary process, a 60 wt % PTFE water dispersion was mixed with PEO and water as follows. First 5 mL of water was added to 1.4 g of PEO. The water and PEO were mixed until the PEO was fully dissolved and the solution created a thick gel. 30 mL of 60 wt % PTFE was then added to the PEO/water mixture. The combined solution was then allowed to sit or mix in a non-agitating jar roller until the solution achieved homogeneity. In other examples, the water, PEO, and PTFE amounts may be controlled to optimize the viscosity, PEO/PTFE ratio, or other properties of the mixture. In some instances adding water to the PEO before mixing with the PTFE dispersion may aid in reducing the number of large solid chunks in the mixture, lower the preparation time for the mixtures, and reduce the time needed for the combined mixture to solubilize.

(64) Membranes composed of electrospun PTFE may have a microstructure composed of many fibers crossing and each other at various and random points. The electrospinning process may control the thickness of this structure and, thereby the relative permeability of the membrane. As more and more strands of PTFE are electrospun onto a membrane, the membrane may both increase in thickness and decrease in permeability (due to successive layers of strands occluding the pores and openings of layers below). (This microstructure is shown in FIGS. 9A-11D which are discussed in more detail below.)

(65) The complex and random microstructure of electrospun PTFE presents a challenge to the direct measurement of the average pore size of the membrane. Average pore size can be indirectly determined by measuring the permeability of the membrane to fluids using known testing techniques and instruments. Once the permeability is determined, that measurement may be used to determine an “effective” pore size of the electrospun PTFE membrane. As used herein, the “pore size” of an electrospun PTFE membrane refers to the pore size of a membrane which corresponds to the permeability of the electrospun PTFE when measured using ASTM standard F316 for the permeability measurement. This standard is described in ASTM publication F316 “Standard Test Methods for Pore Size Characteristics of Membrane Filters by Bubble Point and Mean Flow Pore Test,” which is incorporated herein by reference.

(66) In some applications it may be desirable to create a stent **100** with an outer layer **210** which is substantially impermeable. Such a layer may decrease the incidence of lumen tissue surrounding the stent growing into the stent. This may be desirable in applications where the stent is used to treat stenosis or other occlusions; an impermeable outer layer may prevent tissue from growing into the lumen of the stent and reblocking or restricting the body lumen. In some embodiments a substantially impermeable outer layer may be produced by using electrospun PTFE with an average pore size of about 0 microns to about 1.5 microns. In other embodiments, the impermeable layer may have an average pore size less than about 0.5 microns. In yet other embodiments, the impermeable layer may have an average pore size less than about 1.0 microns. In some embodiments, the impermeable layer may be a layer other than the outer layer, such as a tie layer, an intermediate layer or an inner layer. Furthermore, a substantially impermeable layer may be formed of fluorinated ethylene propylene (FEP) which is applied, for example, as a film or dip coating. Furthermore, FEP may be electrospun with a small average pore size to create a substantially impermeable layer.

(67) In other potential embodiments it may be desirable to create a stent with an outer layer **210** which is more porous. A porous outer layer **210** may permit healing and the integration of the prosthesis into the body. For instance, tissue of the surrounding lumen may grow into the porous outer diameter. This “tissue ingrowth” may permit healing at the therapy site. In some

embodiments a porous outer layer **210** may be formed of electrospun PTFE.

(68) In certain embodiments a relatively porous inner layer **220** may be desirable. This layer may or may not be used in conjunction with a substantially impermeable outer layer **210**. A relatively porous inner layer may permit endothelial grown on the inside diameter of the stent **100** which may be desirable for healing, biocompatibility, and reducing turbulent blood flow within the stent. In some embodiments the inner layer may be comprised of electrospun PTFE with an average pore size of about 1 microns to about 12 microns, such as from about 2 microns to about 8 microns, or from about 3 microns to about 5 microns, or alternatively from about 3.5 to about 4.5 microns.

(69) FIG. **6B** illustrates a cross sectional view of a stent with an outer layer **210**, an inner layer **220**, and a wire scaffold **110**. Additionally, the location between the outer layer **210** and the inner layer **220** is illustrated as **230**. It will be appreciated that in embodiments where there are only two layers, there may not be a gap between the two layers, but the outer layer **210** and inner layer **220** may be in direct contact where they are not separated by the wire **110**.

(70) In other embodiments a third layer may be disposed in the location **230** between the outer layer **210** and the inner layer **220**. In some embodiments this layer may be a “tie layer” configured to promote bonding between the outer layer **210** and the inner layer **220**. In other embodiments the tie layer may further be configured to provide certain properties to the stent as a whole, such as stiffness or tensile strength. Furthermore, in embodiments where both the inner layer **220** and the outer layer **210** are porous in nature, the tie layer may be configured to create an impermeable layer between the two porous layers. In such embodiments the stent may permit cell growth and healing on both the inner and outer surfaces of the stent while still preventing tissue from outside the stent from growing into the lumen and occluding the lumen.

(71) The tie layer may consist of any thermoplastic and may or may not be electrospun. In one embodiment, the tie layer may be expanded PTFE. In another it may be electrospun PTFE. In other embodiments it may be FEP, including electrospun FEP and FEP applied as a film or dip coating. Furthermore, the tie layer may consist of any of the following polymers or any other thermoplastic: dextran, alginates, chitosan, guar gum compounds, starch, polyvinylpyridine compounds, cellulosic compounds, cellulose ether, hydrolyzed polyacrylamides, polyacrylates, polycarboxylates, polyvinyl alcohol, polyethylene oxide, polyethylene glycol, polyethylene imine, polyvinylpyrrolidone, polyacrylic acid, poly(methacrylic acid), poly(itaconic acid), poly(2-hydroxyethyl acrylate), poly(2-(dimethylamino)ethyl methacrylate-co-acrylamide), poly(N-isopropylacrylamide), poly(2-acrylamido-2-methyl-1-propanesulfonic acid), poly(methoxyethylene), poly(vinyl alcohol), poly(vinyl alcohol) 12% acetyl, poly(2,4-dimethyl-6-triazinylethylene), poly(3morpholinylethylene), poly(N-1,2,4-triazolyethylene), poly(vinyl sulfoxide), poly(vinyl amine), poly(N-vinyl pyrrolidone-co-vinyl acetate), poly(g-glutamic acid), poly(Npropanoyliminoethylene), poly(4-amino-sulfo-aniline), poly [N-(p-sulphophenyl)amino-3-hydroxymethyl-1,4phenyleneimino-1,4-phenylene], isopropyl cellulose, hydroxyethyl, hydroxylpropyl cellulose, cellulose acetate, cellulose nitrate, alginic ammonium salts, i-carrageenan, N-[(3'-hydroxy-2',3'-dicarboxy)ethyl]chitosan, konjac glocomannan, pullulan, xanthan gum, poly(allylammonium chloride), poly(allylammonium phosphate), poly(diallyldimethylammonium chloride), poly(benzyltrimethylammonium chloride), poly(dimethyldodecyl(2-acrylamidoethyl) ammonium bromide), poly(4-N-butylpyridiniummethylene iodine), poly(2-N-methylpyridinium methylene iodine), poly(N-methylpyridinium-2,5-diethenylene), polyethylene glycol polymers and copolymers, cellulose ethyl ether, cellulose ethyl hydroxyethyl ether, cellulose methyl hydroxyethyl ether, poly(1-glycerol methacrylate), poly(2-ethyl-2-oxazoline), poly(2-hydroxyethyl methacrylate/methacrylic acid) 90:10, poly(2-hydroxypropyl methacrylate), poly(2-methacryloxyethyltrimethylammonium bromide), poly(2-vinyl-1-methylpyridinium bromide), poly(2-vinylpyridine N-oxide), poly(2-vinylpyridine), poly(3-chloro-2-hydroxypropyl 2-methacryloxyethyltrimethylammonium chloride), poly(4vinylpyridine N-oxide), poly(4-vinylpyridine), poly(acrylamide/2-

methacryloxyethyltrimethylammonium bromide) 80:20, poly(acrylamide/acrylic acid), poly(allylamine hydrochloride), poly(butadiene/maleic acid), poly(diallyldimethylammonium chloride), poly(ethyl acrylate/acrylic acid), poly(ethylene glycol) bis(2-aminoethyl), poly(ethylene glycol) monomethyl ether, poly(ethylene glycol)bisphenol A diglycidyl ether adduct, poly(ethylene oxide-bpropylene oxide), poly(ethylene/acrylic acid) 92:8, poly(llysine hydrobromide), poly(l-lysine hydrobromide), poly(maleic acid), poly(n-butyl acrylate/2methacryloxyethyltrimethylammonium bromide), poly(Niso-propylacrylamide), poly(N-vinylpyrrolidone/2dimethylaminoethyl methacrylate), dimethyl sulfatequaternary, poly(N-vinylpyrrolidone/vinyl acetate), poly(oxyethylene) sorbitan monolaurate (Tween 20®), poly(styrenesulfonic acid), poly(vinyl alcohol), N-methyl-4(4'formylstyryl)pyridinium, methosulfate acetal, poly(vinyl methyl ether), poly(vinylamine) hydrochloride, poly(vinylphosphonic acid), poly(vinylsulfonic acid) sodium salt and polyaniline.

(72) Regardless of the material, the tie layer may or may not be electrospun. Further, in certain embodiments the stent may consist of two or more tie layers. The tie layer may be formed in any manner known in the art and attached to the inner and outer layers in any manner known in the art. For example, the tie layer may comprise a sheet of material which is wrapped around the inner layer **210** or a tube of material which is slipped over the inner layer **210** which is then heat shrunk or otherwise bonded to the inner and outer layers. Further, in embodiments where the tie layer is electrospun, it may be electrospun directly onto the inner layer **210**, the scaffolding, or both. In some instances the tie layer may be melted after the stent is constructed to bond the tie layer to adjacent layers of the stent covering.

(73) Furthermore, tie layers may be configured to change the overall properties of the stent covering. For example, in some instances a cover comprised solely of electrospun PTFE (of the desired pore size) may not have desired tensile or burst strength. A tie layer comprised of a relatively stronger material may be used to reinforce the PTFE inner layer, the PTFE outer layer, or both. For example, in some instances FEP layers may be used to increase the material strength of the cover.

(74) It will also be appreciated that one or more layers of electrospun PTFE may be used in connection with a scaffolding structure other than that disclosed herein. In other words, the disclosure above relating to covers, layers, tie layers, and related components is applicable to any type of scaffolding structure as well as to stents or grafts with no separate scaffolding structure at all.

(75) FIG. 7 illustrates a cross section of a stent **100** disposed within a body lumen **50**. The stent includes wire scaffolding **110** and a cover **200**. In embodiments where the cover **200** is composed of an outer layer and an inner layer, the outer layer may be disposed adjacent to the body lumen while the inner layer may be disposed toward the inside portion of the body lumen. In particular, in embodiments where the stent is not substantially tubular in shape, the outer cover layer may be defined as the layer disposed adjacent the body lumen wall and the inner cover layer defined as the layer disposed toward the inner portion of the body lumen.

(76) In some embodiments, a cover **200** may be formed by electrospinning a membrane onto a spinning mandrel. In other words, the collection device may comprise a mandrel, such as a substantially cylindrical mandrel, which rotates during the electrospinning process. Varying the speed at which the mandrel rotates may influence certain properties of the membrane. For example, in some embodiments, the density of the membrane (and thereby the average pore size) may be related to the rotational speed of the mandrel. Further, the directionality of the fibers, or the degree to which the fibers are deposited in a more controlled direction or manner, may be related to the rotational speed of the mandrel. In some instances a collection mandrel may rotate at rates between about 1 RPM and about 500 RPM during the electrospinning process, including rates from about 1 RPM to about 50 RPM or at about 25 RPM. A membrane of electrospun PTFE formed onto a spinning mandrel may thus comprise a tubular membrane having no seam and substantially

isotropic properties.

(77) Once a membrane has been electrospun onto a mandrel, the membrane may then be sintered. In the case of PTFE, the membrane may be sintered at temperatures of about 385 degrees C., including temperatures from about 360 degrees C. to about 400 degrees C. Sintering may tend to set the structure of the PTFE, meaning sintering reduces the softness or flowability of the PTFE. Furthermore, sintering may evaporate any water or PEO mixed with the PTFE, resulting in a material comprised substantially of pure PTFE.

(78) In some embodiments, a PTFE layer may be spun onto a mandrel and then sintered. Once the membrane is sintered, the tube of material may be removed from the mandrel, then slid back on the mandrel (to initially break any adherence of the membrane to the mandrel). In other instances, low friction coatings may alternatively or additionally be applied to the mandrel before the membrane is electrospun. Once the membrane is reapplied to the mandrel, a wire scaffolding can be formed over the mandrel and the membrane. A second layer of material may then be spun onto the scaffolding and the membrane, and subsequently sintered. Additionally layers may also be added.

(79) In some instances, the layers may comprise a first layer of PTFE, a second layer of FEP, and a third layer of PTFE. The properties of each of these layers, including average pore size, may be controlled to form coating that inhibit growth of tissue through a particular layer or that permits endothelial growth on a particular layer.

(80) In another example, a first layer of PTFE may be spun on a mandrel, sintered, removed from the mandrel, replaced and the mandrel, and a scaffolding structure applied. An FEP layer may then be applied by dipping, spraying, application of a film layer, electrospinning, or other processing. The FEP layer may or may not be sintered before applying an outer PTFE layer.

(81) In another particular example, a first layer of PTFE may again be spun on a mandrel, sintered, removed, replaced, and a scaffolding structure applied. An FEP layer may then be applied as a film layer. In some instances it may be “tacked” into place, for example by a soldering iron. An outer tube of PTFE (which may be formed separately by electrospinning onto a mandrel and sintering) may then be disposed over the FEP film layer. The entire construct may then be pressured, for example by applying a compression wrap. In some embodiments this wrap may comprise any suitable material, including a PTFE based material. In other embodiments a Kapton film may be wrapped around the construct before the compression wrap, to prevent the construct from adhering to the compression wrap.

(82) The compressed layers may then be heated above the melting temperature of the FEP, but below the sintering temperature of the PTFE. For example, the melt temperature of the FEP may be from about 300 degrees C. to about 330 degrees C., including about 325 degrees C. PTFE may be sintered at temperatures from about 360 degrees C. to about 400 degrees C. Thus, the entire construct may be heated to an appropriate temperature such as about 325 degrees C. In some embodiments the construct may be held at this temperature for about 15 to about 20 minutes. This may allow the FEP to “flow” into the porous PTFE nanofiber layers surrounding the FEP. The joining of the FEP tie layer to the PTFE outer and inner cover layers may increase the strength of the finished covering. The construct may then be cooled and the compression wrap and the Kapton film discarded. The construct may then be removed from the mandrel.

(83) A stent formed by the exemplary process described above may be configured with desired characteristics of porosity and strength. In some instances the FEP material may coat the PTFE nanofibers, but still allow for porosity which permits endothelial growth. The degree to which the FEP coats the PTFE may be controlled by the temperature and time of processing. The lower the temperature and/or the shorter the time the construct is held at temperature, the less the FEP may flow. In some instances a tie layer of FEP which is impervious the tissue growth through the layer may be formed by heating the construction only to about 260 degrees C.

(84) Additionally, in some embodiments a stent may also include a cuff at one or both ends of the stent. The cuff may be an additional coating on the outside diameter of the stent, disposed adjacent

one of the ends of the stent. The cuff may be configured to promote rapid cellular ingrowth into the cuff; for example the cuff may be more porous than the outer layer of the covering of the stent. Factors such as porosity, type of coating, type of material, use of organic material, and/or use of composite materials formed of synthetic material and organic material may be used to create a cuff configured for rapid tissue ingrowth. Again, the cuff may be configured to promote rapid growth or endothelialization at one or both ends of the stent. In some embodiments cuffs may be disposed adjacent both ends of the stent. The cuff or cuffs may tend to “anchor” the ends of the stent with respect to the vessel walls, reducing the relative movement of the stent ends with respect to the vessel walls. Such a reduction in movement may lessen irritation of the vessel by the stent ends, minimizing complications such as stenosis. Cuffs may be configured for use in CVO type applications in some instances.

(85) In some embodiments, the outer layer of the covering of the stent may be relatively non-porous to limit tissue growth through the layer, but the cuff, disposed about the outer cover layer may provide a section near each end at which some ingrowth may occur.

(86) The cuff may be comprised of an electrospun material, such as PTFE, and may be bonded to the outer covering layer through any method, including methods described herein. For example, a layer of FEP may be disposed between the outer covering layer and the cuff and heated to bond the layers. In other embodiments the cuff may comprise a collagen layer which is glued to the stent. Further, a co-electrospun collagen and PTFE cuff may be utilized.

(87) FIGS. 8A-9D are scanning electron micrograph (SEM) images of an exemplary embodiment of a stent covering. FIGS. 8A-8D are images of the outer layer of the covering while FIGS. 9A-9D are images of the inner layer of the covering. For each SEM, the electrospun PTFE was covered with a very thin layer of gold in order to make the structure visible on an SEM image.

(88) FIG. 8A is an SEM image of the outer covering at 750× magnification, FIG. 8B an SEM image at 1500× magnification, and FIGS. 8C and 8D at 3000× magnification. Similarly, FIG. 9A is an image of the inner covering at 750× magnification, FIG. 9B at 1500× magnification, and FIGS. 9C and 9D at 3000× magnification.

(89) These SEM images reflect the microstructure of electrospun PTFE, depicting the randomly deposited criss-crossing branches of PTFE that form the covering.

(90) FIGS. 10A-11D are scanning electron micrograph (SEM) images of a second exemplary embodiment of a stent covering. FIGS. 10A-10D are images of the outer layer of the covering while FIGS. 11A-11D are images of the inner layer of the covering. Again, for each SEM, the electrospun PTFE was covered with a very thin layer of gold in order to make the structure visible on an SEM image.

(91) FIG. 10A is an SEM image of the outer covering at 750× magnification, FIG. 10B an SEM image at 1500× magnification, and FIGS. 10C and 10D at 3000× magnification. Similarly, FIG. 11A is an image of the inner covering at 750× magnification, FIG. 11B at 1500× magnification, and FIGS. 11C and 11D at 3000× magnification.

(92) While specific embodiments of stents have been illustrated and described, it is to be understood that the disclosure provided is not limited to the precise configuration and components disclosed. Various modifications, changes, and variations apparent to those of skill in the art having the benefit of this disclosure may be made in the arrangement, operation, and details of the methods and systems disclosed, with the aid of the present disclosure.

(93) Without further elaboration, it is believed that one skilled in the art can use the preceding description to utilize the present disclosure to its fullest extent. The examples and embodiments disclosed herein are to be construed as merely illustrative and exemplary and not a limitation of the scope of the present disclosure in any way. It will be apparent to those having skill, having the benefit of this disclosure, in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the disclosure herein.

Claims

1. A vascular prosthesis, comprising: a scaffolding structure configured to resist radial compression when disposed in a lumen of a patient, and a coating disposed on at least a portion of the scaffolding structure, the coating comprising a first layer of polytetrafluoroethylene (PTFE), a non-porous tie layer coupled to the first layer, and a second layer of porous PTFE, wherein the non-porous tie layer is impervious to tissue growth into the non-porous tie layer, and wherein portions of the tie layer are disposed within pores of the second layer.
2. The vascular prosthesis of claim 1, wherein: the first layer comprises serially deposited PTFE fibers.
3. The vascular prosthesis of claim 2, wherein the serially deposited PTFE fibers are electrospun PTFE fibers.
4. The vascular prosthesis of claim 1, wherein the first layer is permeable to tissue growth into the first layer.
5. The vascular prosthesis of claim 1, wherein the first layer defines a luminal surface of the vascular prosthesis.
6. The vascular prosthesis of claim 1, wherein the tie layer is non-fibrous.
7. The vascular prosthesis of claim 2, wherein the second layer is permeable to tissue growth into the second layer.
8. The vascular prosthesis of claim 2, wherein the second layer comprises serially deposited PTFE fibers.
9. The vascular prosthesis of claim 2, wherein the second layer comprises electrospun PTFE fibers.
10. A medical appliance, comprising: a scaffolding structure comprising a single wire; and a coating disposed on at least a portion of the scaffolding structure, the coating comprising: a first layer of electrospun PTFE fibers; a tie layer of fluorinated ethylene propylene (FEP) film coupled to the first layer; and a second layer of porous PTFE, wherein the tie layer is non-porous, and wherein portions of the tie layer is disposed within pores of the second layer of porous PTFE.
11. The medical appliance of claim 10, wherein the first layer defines a luminal surface of the medical appliance.
12. The medical appliance of claim 10, wherein the first layer is permeable to tissue growth into the first layer.
13. The medical appliance of claim 10, wherein the tie layer is impervious to tissue growth into the tie layer.
14. The medical appliance of claim 10, wherein the tie layer is non-fibrous.
15. The medical appliance of claim 10, wherein the second layer comprises electrospun PTFE fibers.
16. A medical stent, comprising: a scaffolding structure comprising a wave configuration; and a coating disposed on at least a portion of the scaffolding structure, the coating comprising: an inner layer of electrospun PTFE fibers; a tie layer of FEP film coupled to the inner layer and impervious to tissue growth into the tie layer; and an outer layer of electrospun PTFE fibers, wherein the tie layer is non-porous, and wherein portions of the tie layer is disposed within pores of the outer layer of electrospun PTFE.
17. The medical stent of claim 16, wherein the inner layer defines a luminal surface of the medical stent.
18. The medical stent of claim 16, wherein the inner layer is permeable to tissue growth into the inner layer.
19. The medical stent of claim 16, wherein the tie layer is non-fibrous.

20. The medical stent of claim 16, wherein the outer layer is permeable to tissue growth into the outer layer.
